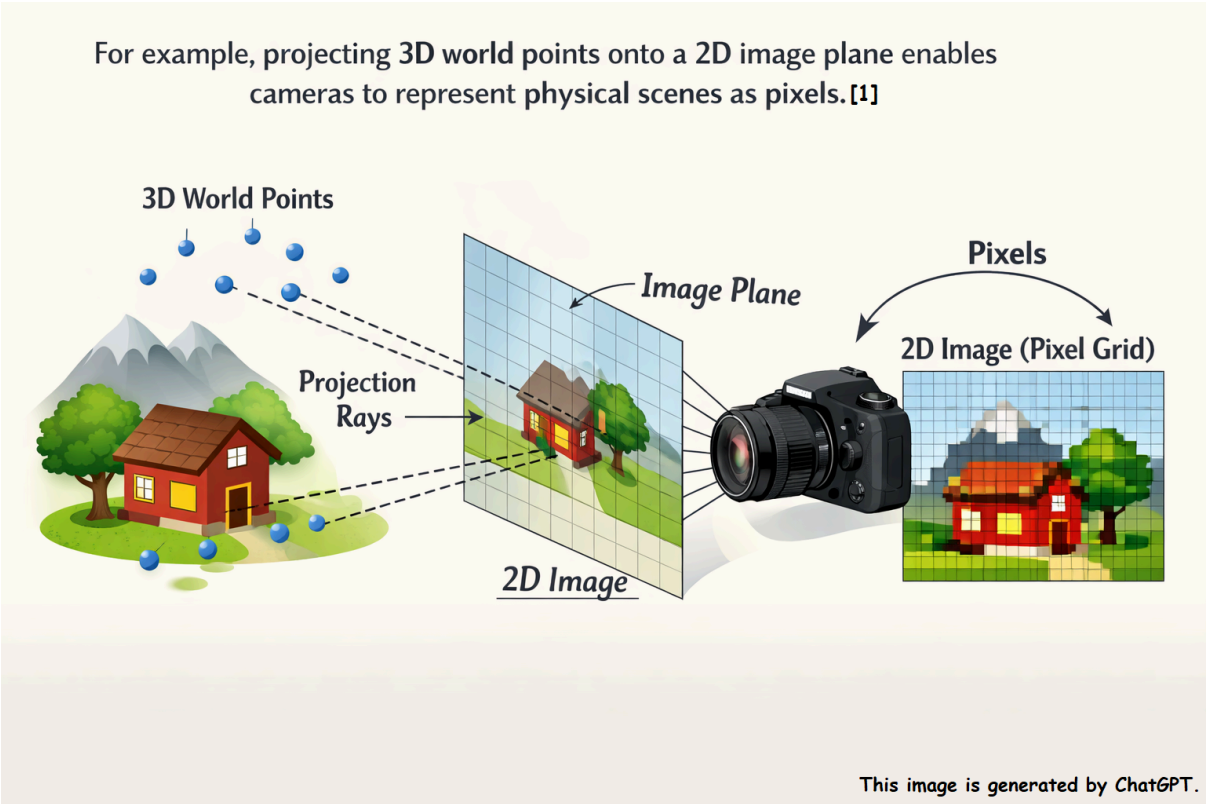
AI Capstone
Homework 1:
BEV Projection
Announce: 3/9, Deadline: 3/23 23:59

Introduction

Coordinate transformations are fundamental to how computers understand and operate in space. They allow data expressed in one coordinate system to be accurately represented in another, making it possible to relate different perspectives, sensors, or reference frames through operations such as rotation, translation, and projection.



In robotics, transformations connect the robot frame, sensor frame, and world frame, allowing consistent localization and mapping. They also form the basis for computing forward and inverse kinematics. This concept can be further understood by referring to the following resource [2]. The implementation of this transformation process will be covered later in HW4. In computer graphics, objects defined in local coordinates must be transformed into world and screen

coordinates before rendering. Across these domains, coordinate transformations are the mathematical foundation that ensures spatial consistency, data fusion, and meaningful interpretation of geometric information.

In this assignment, you will use images collected in **Habitat** to implement a coordinate transformation that projects **four** selected points from a **bird's-eye view (BEV)**, selected in either clockwise or counterclockwise order, onto the **front-view** perspective.

GitHub repository:
https://github.com/HCIS-Lab/ai-capstone26/tree/main/hw1

Implementation

We provide two pairs of images (first and second floor) of front view and top view in the ***bev_data*** folder. The positions and orientations of these two pinhole cameras [3] are different. The following links provide a tutorial on the pinhole camera model [4] [5].

You need to calculate the relative pose of two cameras to complete the task.

- **Camera1 (front): position (0, 1, 0) orientation (0, 0, 0)**
- **Camera2 (bev): position (0, 2.5, 0) orientation (-(np.pi/2), 0, 0)**

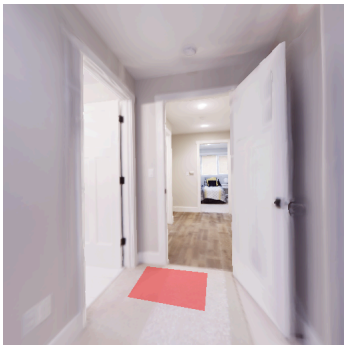
Projection Procedure

1. Load the BEV (top-view) image.
2. Select **four** ground-plane points on the BEV image in either clockwise or counterclockwise order using the mouse.
3. Define a closed region based on the selected **four** points.
4. Project the defined region from the BEV image onto the corresponding perspective (front-view) image.
5. Save the resulting projected image as a *.png* file.

- **For intrinsic parameters, an agent camera utilizes a pinhole camera model (the resolution of an image is 512 x 512)**
- **Horizontal and vertical FOV are 90 degrees**
- **The principal point is assumed to be located at the center of the image**

- **Restriction: Do not import or use any additional Python packages. Only the libraries already included in bev2front.py (*cv2* and *numpy*) are permitted.**
- **We have already provided a code template in bev2front.py; you only need to complete the coordinate transformation logic in that file.**

Example Results



bev1.png front1.png bev2.png front2.png

Grading

Report 100%

Your report should be **written in English**, be concise, and include the following content:

1. Implementation (70%)
    a. Code
    Detailed explanation of your implementation. For example, how you do the projection.
    b. Result and Discussion
        i. Result of your projection (2 different pairs). Like the example result above.
        ii. Anything you want to discuss.
        iii. Any reference you take.

2. Questions (30%)
    a. How do the intrinsic and extrinsic matrices contribute to the BEV-to-front-view projection?
    b. What factors may cause projection misalignment? List at least two factors and explain the reason.

QA Page

If you have any questions about this homework, please ask them on the following Notion page. We will answer them as soon as possible. Additionally, we encourage you to answer other students' questions if you can.

https://www.notion.so/Homework-I-QA-Page-312c9908e8868024a7bdc247db77abc5

Submission

Due Date: 2026/3/23 23:59

Please compress all your code files and the report (.pdf) into a single file named `{STUDENT ID}_hw1.zip` and submit it to the New E3 System.

ex : 123456789_hw1.zip

The file structure should look like:

```
📂{student_id}_hw1.zip
 ├ 📜 bev2front.py
 ├ 📜 report.pdf
 └ 📜 README.md      #  describe how to run your program
```

- **Wrong submission format leads to -10 points.**
- **Late submission leads to -20 points per day.**
- **Plagiarism is strictly prohibited, including assignments written on GitHub. If we find any instances, it will result in a zero score without any room for objection.**
- **Reports not written in English will receive a score of zero.**

References

[1] Computer Vision: The Camera Matrix

[2] Solved Example - Forward Kinematics

[3] Pinhole camera model

[4] Lecture 8: *Camera Models*, by Juan Carlos Niebles, Stanford AI Lab, from the course by Fei-Fei Li, Stanford Vision Lab.

[5] The Pinhole Camera, Universiteit van Amsterdam

[6] A short derivation to basic rotation around the x-, y- or z-axis

[7] Lie Groups for 2D and 3D Transformations